

Eigenvalues & Eigenvectors

Geometric Algorithms

Practice Problem

Suppose A is a 234×300 matrix. What is the smallest possible value for $\dim(\text{Nul}(A))$? What is the largest possible value?

What is the smallest possible value for $\text{rank}(A)$? What is the largest possible value?

Answer

$$A \in \mathbb{R}^{3 \times 5}$$

$$\dim(\text{Nul}(A)) \leq 5$$

$$\dim(\text{Nul}(A)) \geq 2$$

$$\text{rank}(A) \geq 0$$

$$\text{rank}(A) \leq 3$$

$$\text{rank}(A) + \dim(\text{Nul}(A)) = 5$$

$$300$$

$$T(\vec{r}) = \vec{0}$$

$$300 - 234 = 66$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

$$0$$

$$234$$

Objectives

- » Motivate and introduce the fundamental notion of eigenvalues and eigenvectors
- » Determine how to **verify** eigenvalues and eigenvectors
- » Look at the **subspace** generated by eigenvectors
- » Apply the study of eigenvectors to **dynamical linear systems**

Keyword

Eigenvalues

Eigenvectors

Null Space

Eigenspace

Linear Dynamical Systems

Closed-Form Solutions

Motivation

demo

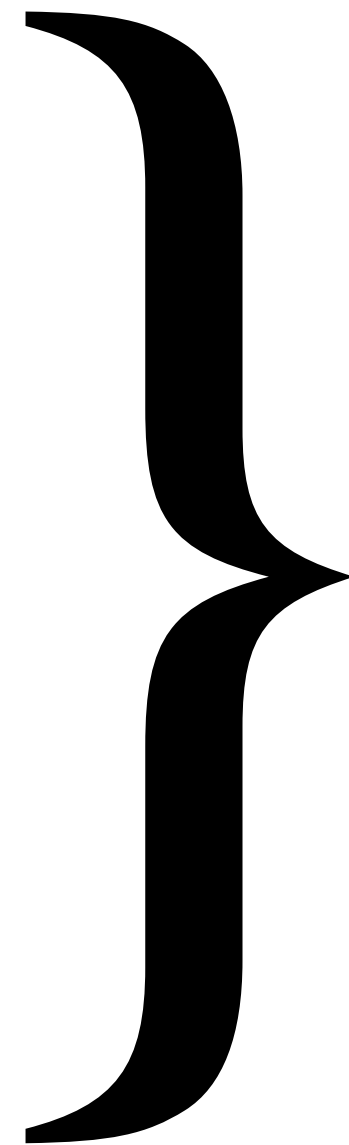
How can matrices transform^{*} vectors?

- » rotations
- » projections
- » shearing
- » reflection
- » scaling/stretching
- » . . .

* square matrices

How can matrices transform^{*} vectors?

- » rotations
- » projections
- » shearing
- » reflection
- » scaling/stretching
- » ...



All matrices do
some combination
of these things

* square matrices

How can matrices transform* vectors?

- » rotations
- » projections
- » shearing
- » reflection
- » scaling/stretching
- » ... **Today's focus**

} All matrices do
some combination
of these things

What's special about scaling?

What's special about scaling?

We don't need a matrix, it's just $\mathbf{x} \mapsto c\mathbf{x}$

If $A\mathbf{v} = c\mathbf{v}$ then it's "easy to describe" what A does to $\mathbf{v}$

Eigenvectors (Informal)

$$A \mathbf{v} = \lambda \mathbf{v}$$

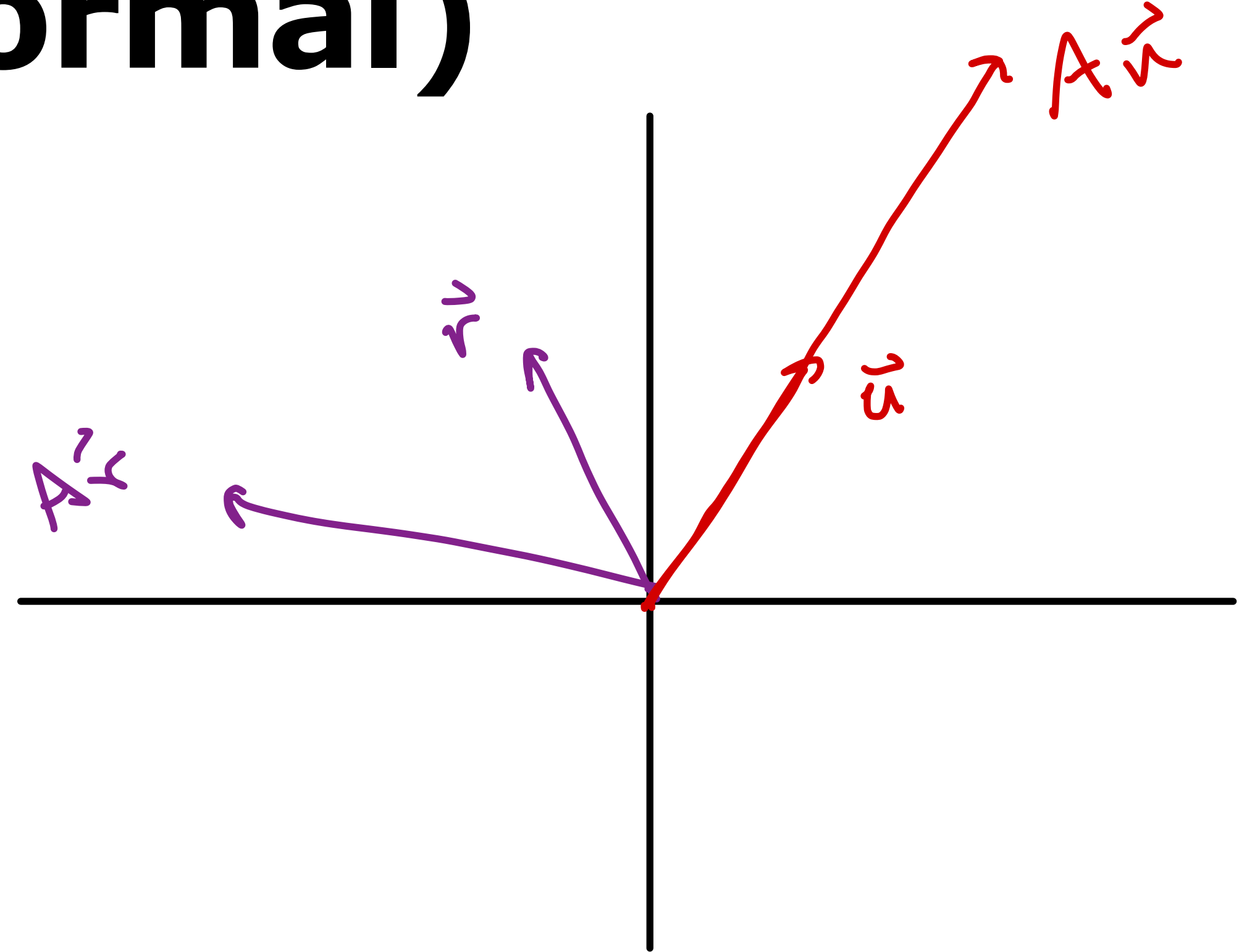
eigenvalue

eigenvector

Eigenvectors (Informal)

$$A \mathbf{v} = \lambda \mathbf{v}$$

Diagram illustrating the eigenvalue equation $A \mathbf{v} = \lambda \mathbf{v}$. The matrix A is shown in black. The vector $\mathbf{v}$ is highlighted in a light blue box and labeled "eigenvector" in blue text below it. The scalar λ is highlighted in a light green box and labeled "eigenvalue" in green text above it.



Eigenvectors of A are stretched by A without changing their direction

Eigenvectors (Informal)

$$A \mathbf{v} = \lambda \mathbf{v}$$

The diagram illustrates the equation $A \mathbf{v} = \lambda \mathbf{v}$. The matrix A is black. The vector $\mathbf{v}$ is blue and highlighted with a light blue square. The equals sign is black. The scalar λ is green and highlighted with a light green square. The vector $\mathbf{v}$ is blue and highlighted with a light blue square. The word "eigenvalue" is written in green above the λ . The word "eigenvector" is written in blue below the second $\mathbf{v}$.

Eigenvectors of A are stretched by A without changing their direction

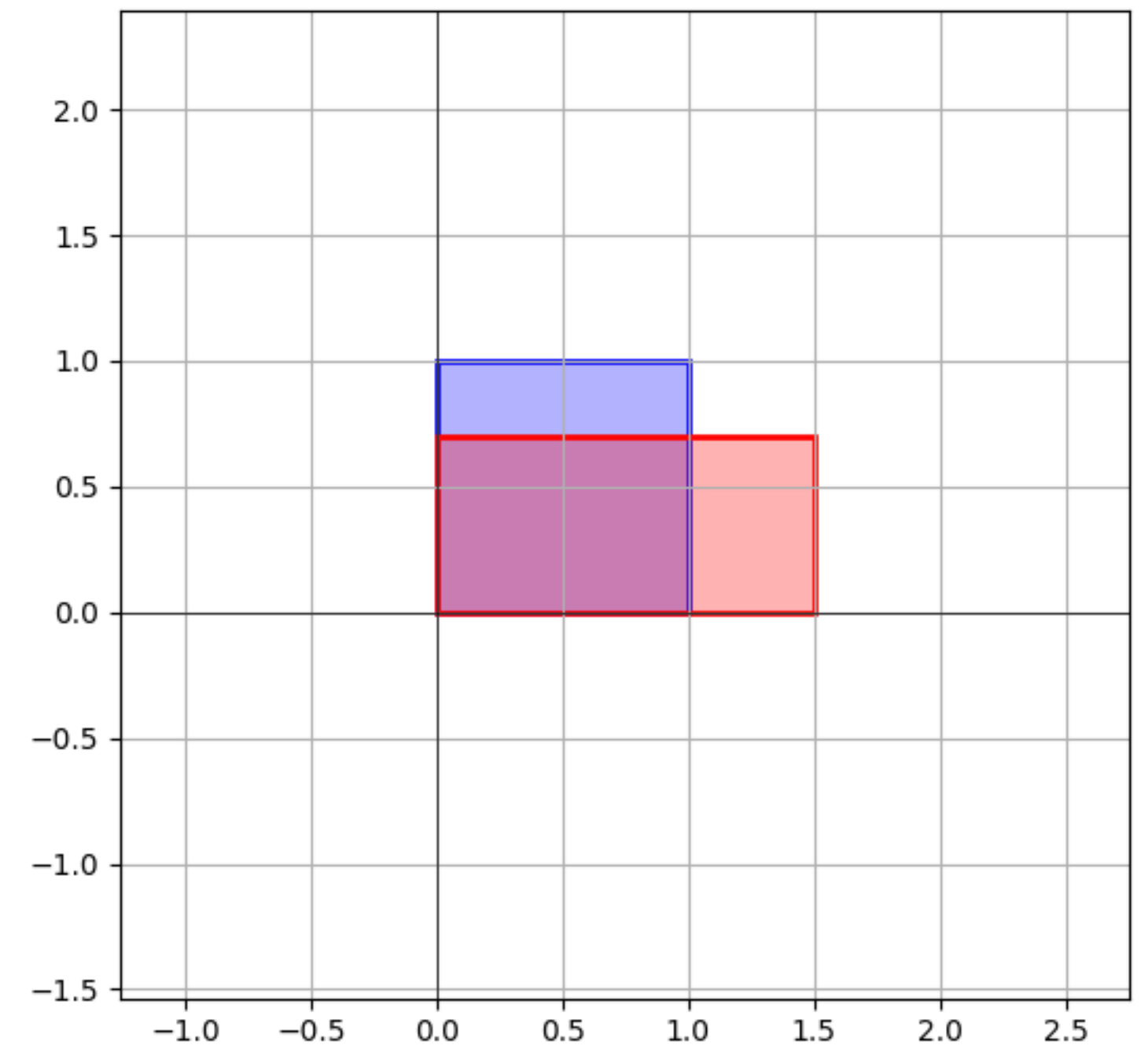
The amount they are stretched is called the **eigenvalue**

Example: Unequal Scaling

$$\begin{bmatrix} 1.5 & 0 \\ 0 & 0.7 \end{bmatrix}$$

It's "easy to describe" how unequal scaling transforms vectors. *It transforms each entry individually and then combines them*

$$\begin{bmatrix} 1.5 & 0 \\ 0 & 0.7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1.5x_1 \\ 0.7x_2 \end{bmatrix}$$
$$= 1.5 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 0.7 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



Eigenbases (Informal)

Eigenbases (Informal)

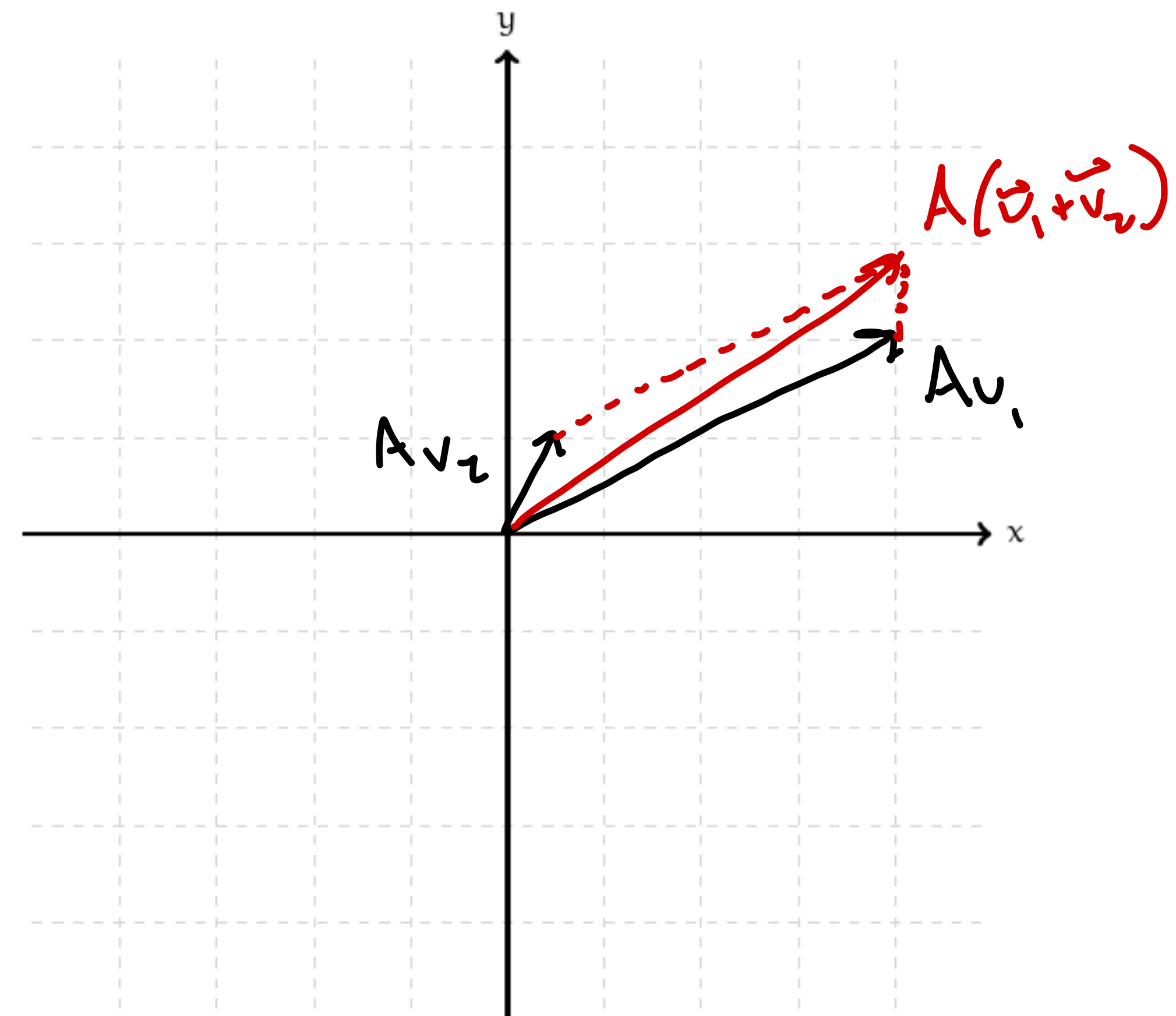
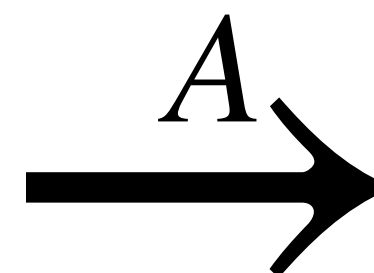
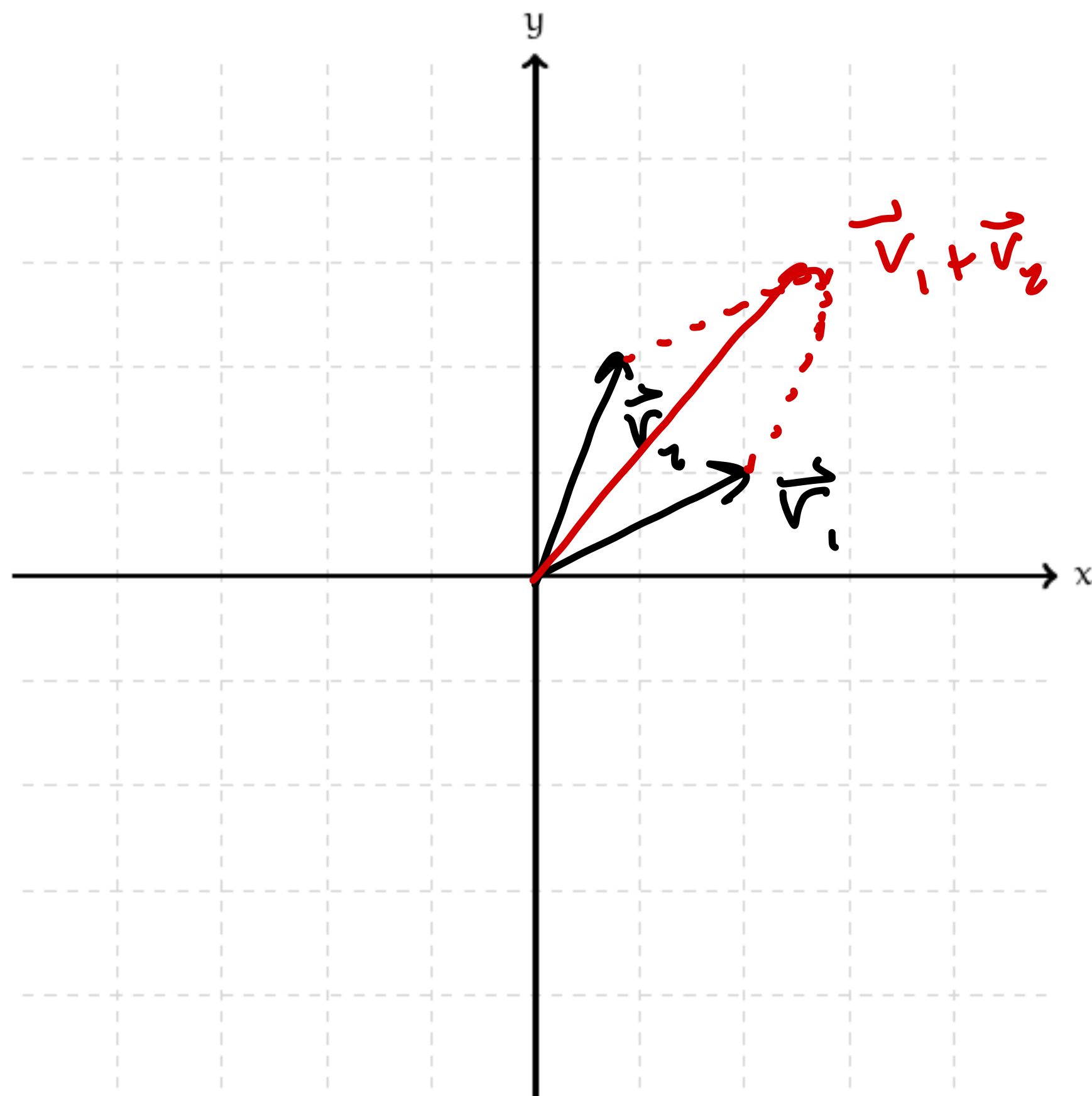
Suppose $\mathbf{v} = 2\mathbf{b}_1 - \mathbf{b}_2 - 5\mathbf{b}_3$ and $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$ are *eigenvectors* of A

It's "easy to describe" how A transforms $\mathbf{v}$:

$$\begin{aligned} A \vec{v} &= A(2\vec{b}_1 - \vec{b}_2 - 5\vec{b}_3) \\ &= 2A\mathbf{b}_1 - A\mathbf{b}_2 - 5A\mathbf{b}_3 \\ &= 2\underline{\lambda_1}\mathbf{b}_1 - \underline{\lambda_2}\mathbf{b}_2 - 5\underline{\lambda_3}\mathbf{b}_3 \end{aligned}$$

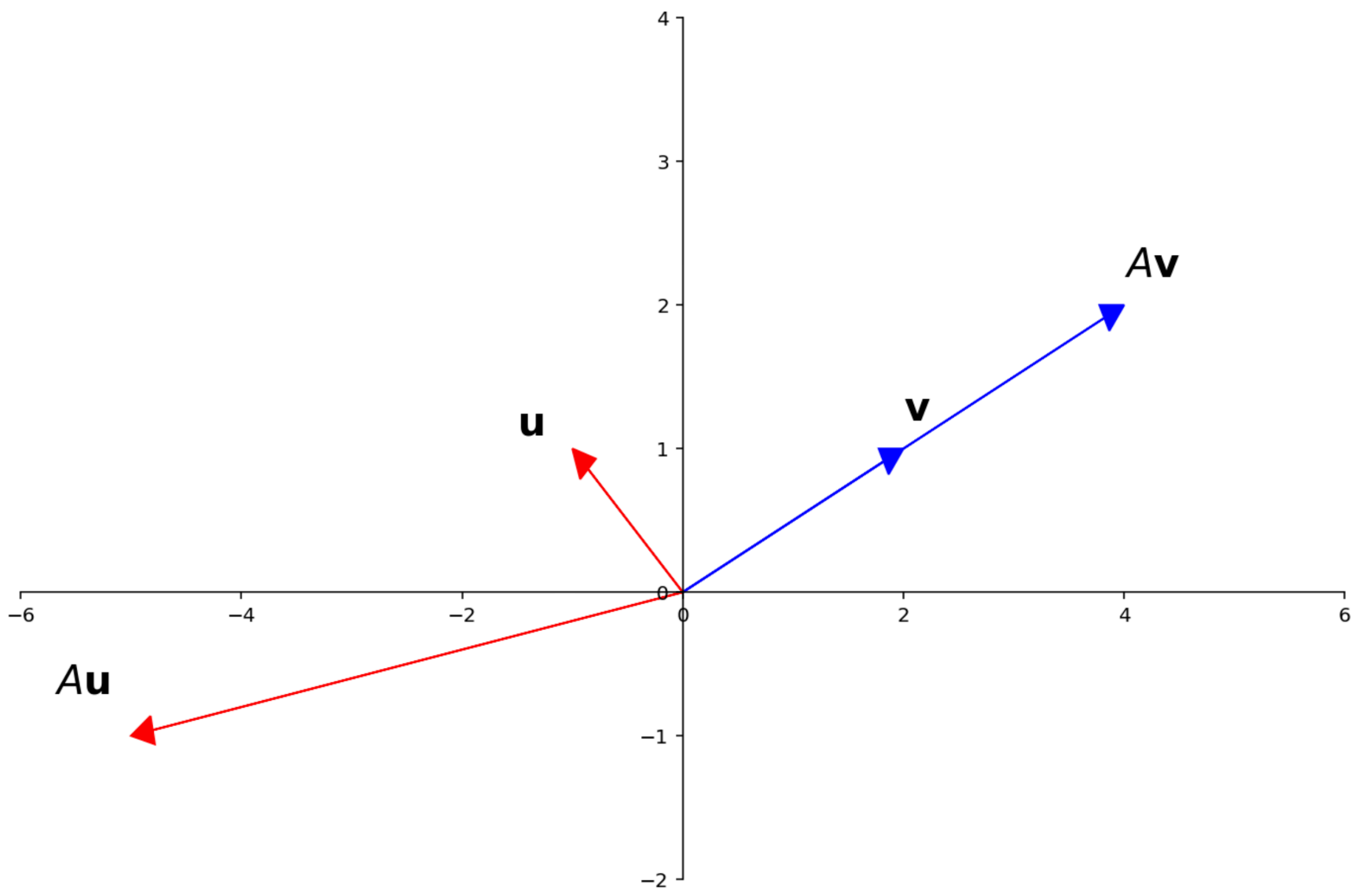
eigenvalues

Eigenbases (Pictorially)

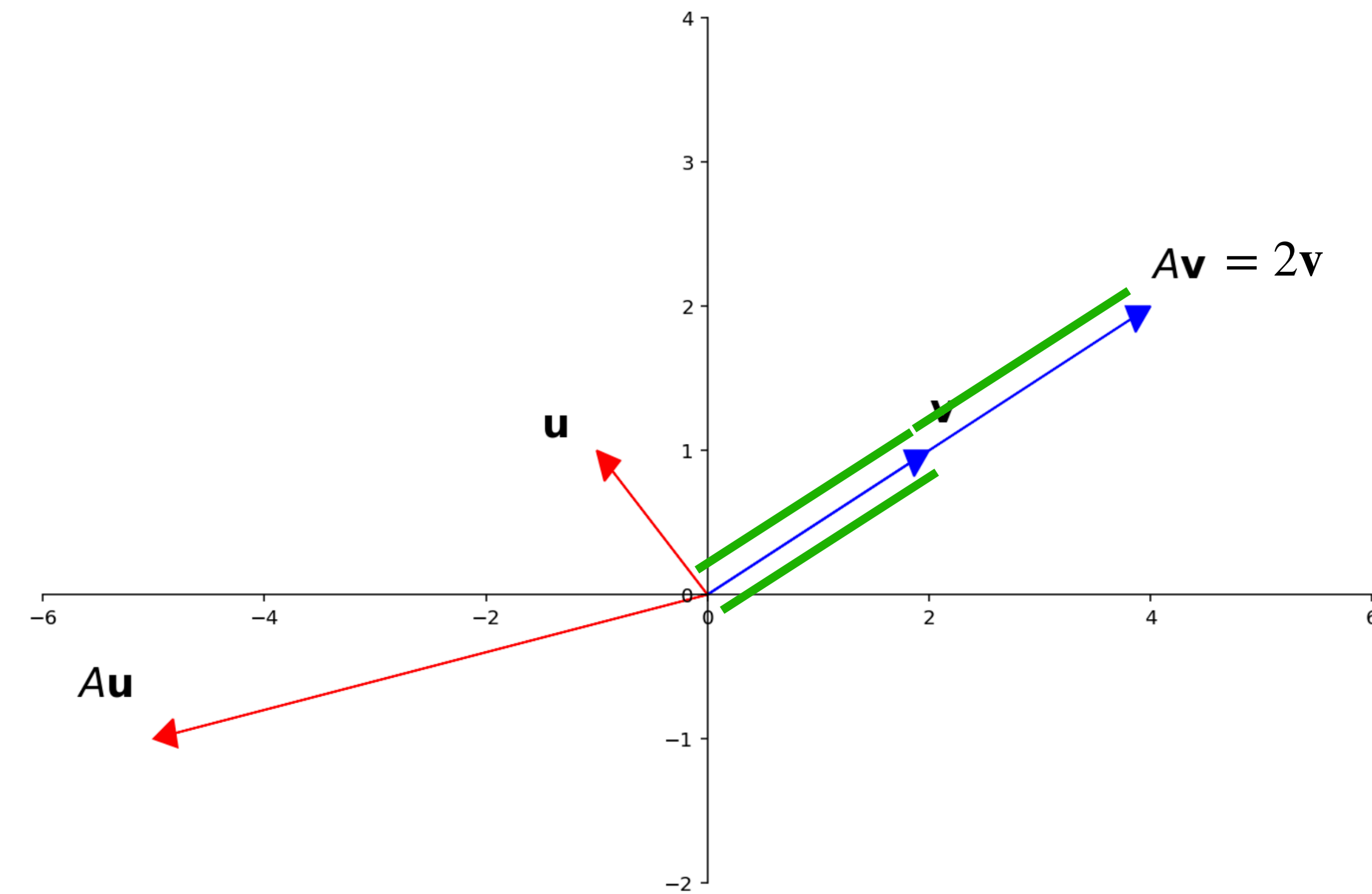


Eigenvalues and Eigenvectors

Formal Definition

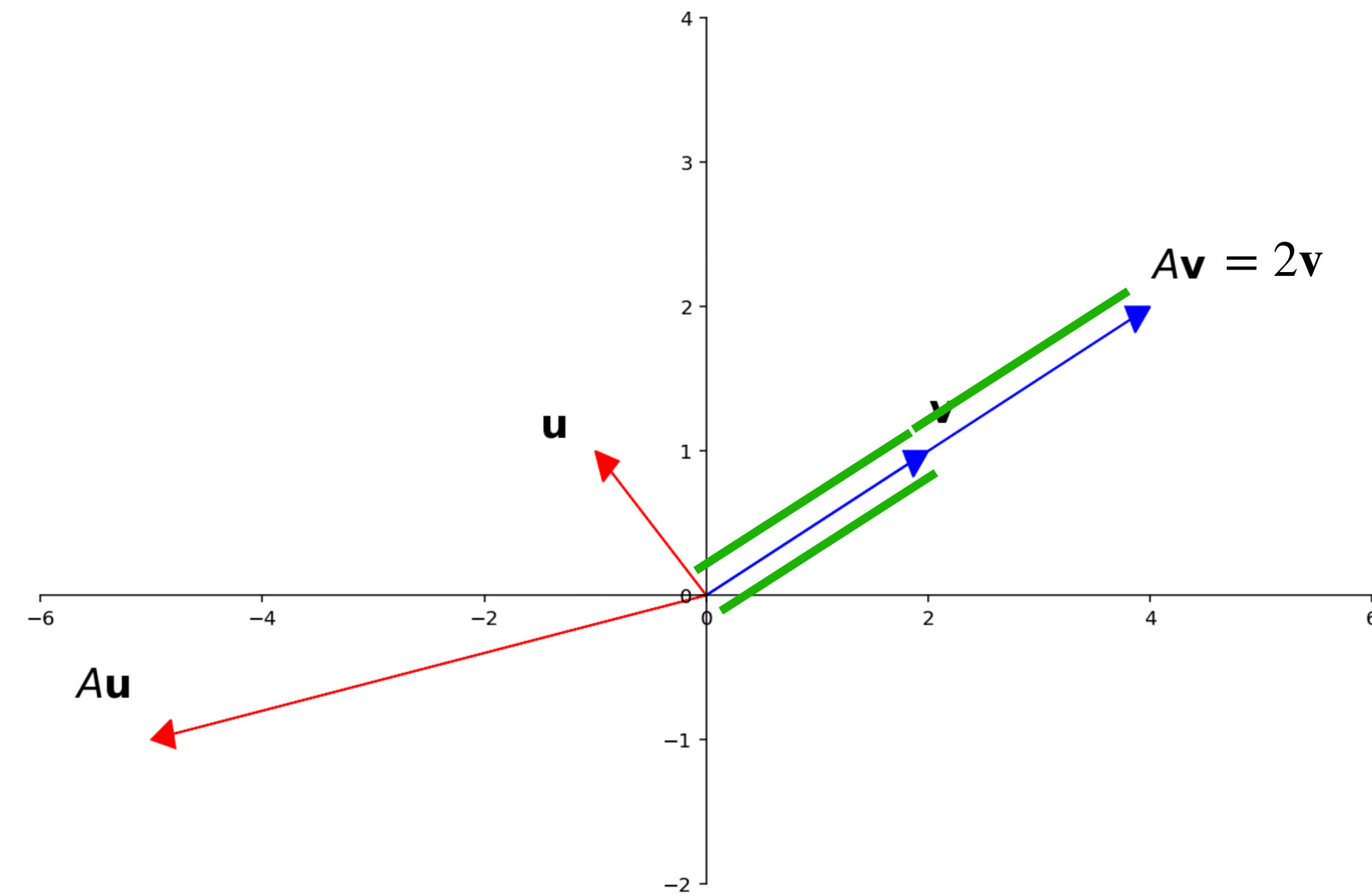


Formal Definition



Def. A *nonzero* vector $\mathbf{v} \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$ are an **eigenvector** and **eigenvalue** for a $A \in \mathbb{R}^{n \times n}$ if $A\mathbf{v} = \lambda\mathbf{v}$

Formal Definition

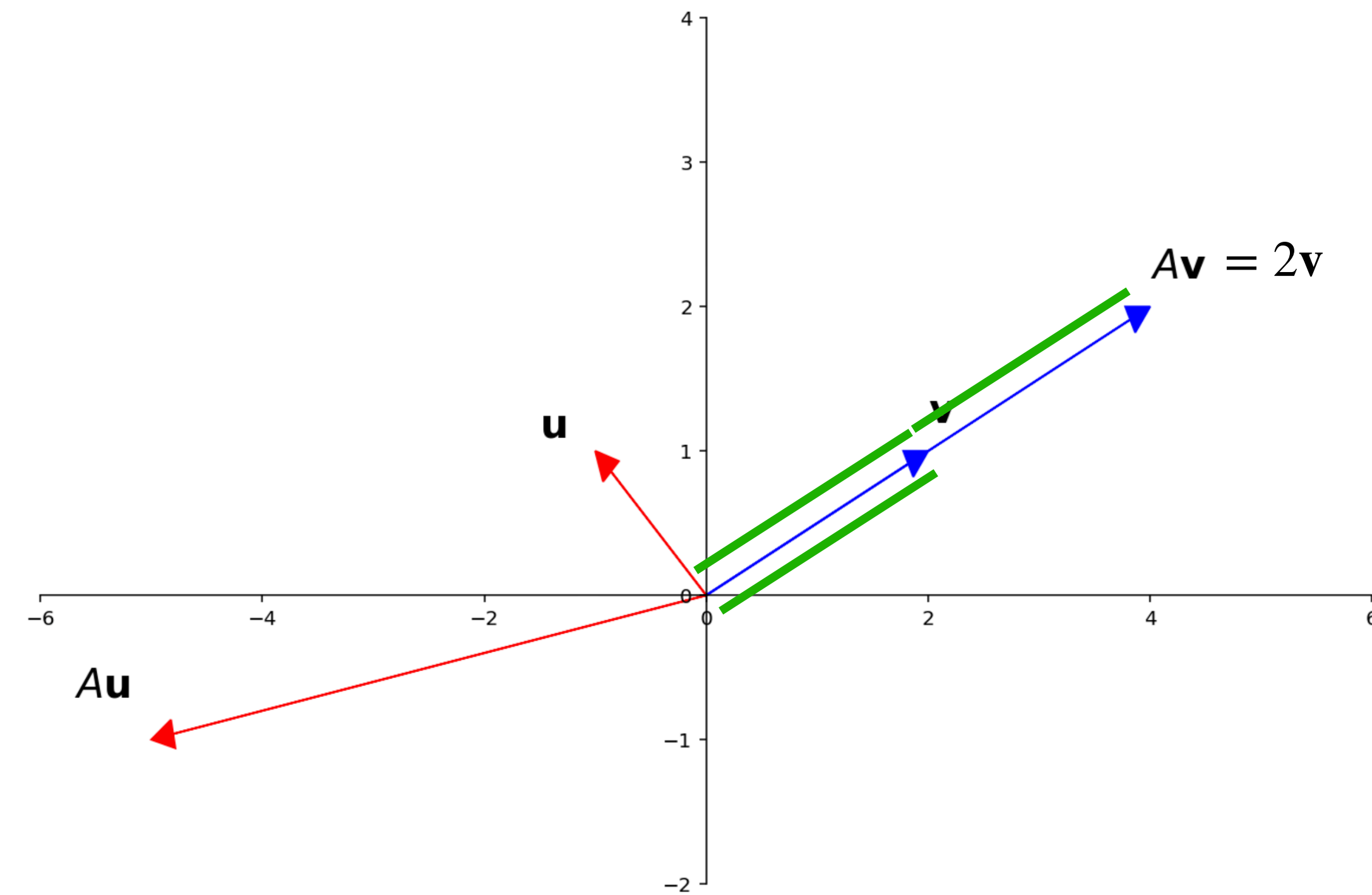


Def. A *nonzero* vector $\mathbf{v} \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$ are an **eigenvector** and **eigenvalue** for a $A \in \mathbb{R}^{n \times n}$ if $A\mathbf{v} = \lambda\mathbf{v}$

We will say that $\mathbf{v}$ is an eigenvector *of/for* the eigenvalue λ , and that λ is the eigenvalue *of/corresponding to* $\mathbf{v}$

Formal Definition

$$\cancel{A \vec{0} = \vec{0} = \lambda \vec{0}}$$



Def. A *nonzero* vector $\mathbf{v} \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$ are an **eigenvector** and **eigenvalue** for a $A \in \mathbb{R}^{n \times n}$ if $A\mathbf{v} = \lambda\mathbf{v}$

We will say that $\mathbf{v}$ is an eigenvector *of/for* the eigenvalue λ , and that λ is the eigenvalue *of/corresponding to* $\mathbf{v}$

Note. Eigenvectors *must* be nonzero, but 0 can be an eigenvalue

What if 0 is an eigenvalue?

$$A \vec{v} = 0 \vec{v} = \vec{0}$$

↑
homogen. lin. eq.

What if 0 is an eigenvalue?

» $\mathbf{v} \in \text{Nul}(A)$

» $\mathbf{v}$ is a nontrivial solution to $A\mathbf{v} = \mathbf{0}$

Extending the IMT (Again)

Extending the IMT (Again)

Thm. A matrix $A \in \mathbb{R}^{n \times n}$ is invertible if and only if it *does not* have 0 as an eigenvalue

Extending the IMT (Again)

Thm. A matrix $A \in \mathbb{R}^{n \times n}$ is invertible if and only if it *does not* have 0 as an eigenvalue

To reiterate, having eigenvalue 0 is equivalent to:

Extending the IMT (Again)

Thm. A matrix $A \in \mathbb{R}^{n \times n}$ is invertible if and only if it *does not* have 0 as an eigenvalue

To reiterate, having eigenvalue 0 is equivalent to:

» $A\mathbf{x} = \mathbf{0}$ has no nontrivial solutions

Extending the IMT (Again)

Thm. A matrix $A \in \mathbb{R}^{n \times n}$ is invertible if and only if it *does not* have 0 as an eigenvalue

To reiterate, having eigenvalue 0 is equivalent to:

- » $A\mathbf{x} = \mathbf{0}$ has no nontrivial solutions
- » the columns of A are linearly dependent

Extending the IMT (Again)

Thm. A matrix $A \in \mathbb{R}^{n \times n}$ is invertible if and only if it *does not* have 0 as an eigenvalue

To reiterate, having eigenvalue 0 is equivalent to:

- » $A\mathbf{x} = \mathbf{0}$ has no nontrivial solutions
- » the columns of A are linearly dependent
- » $\text{Col}(A) \neq \mathbb{R}^n$

Extending the IMT (Again)

Thm. A matrix $A \in \mathbb{R}^{n \times n}$ is invertible if and only if it *does not* have 0 as an eigenvalue

To reiterate, having eigenvalue 0 is equivalent to:

- » $A\mathbf{x} = \mathbf{0}$ has no nontrivial solutions
- » the columns of A are linearly dependent
- » $\text{Col}(A) \neq \mathbb{R}^n$
- » ...

Example: Unequal Scaling

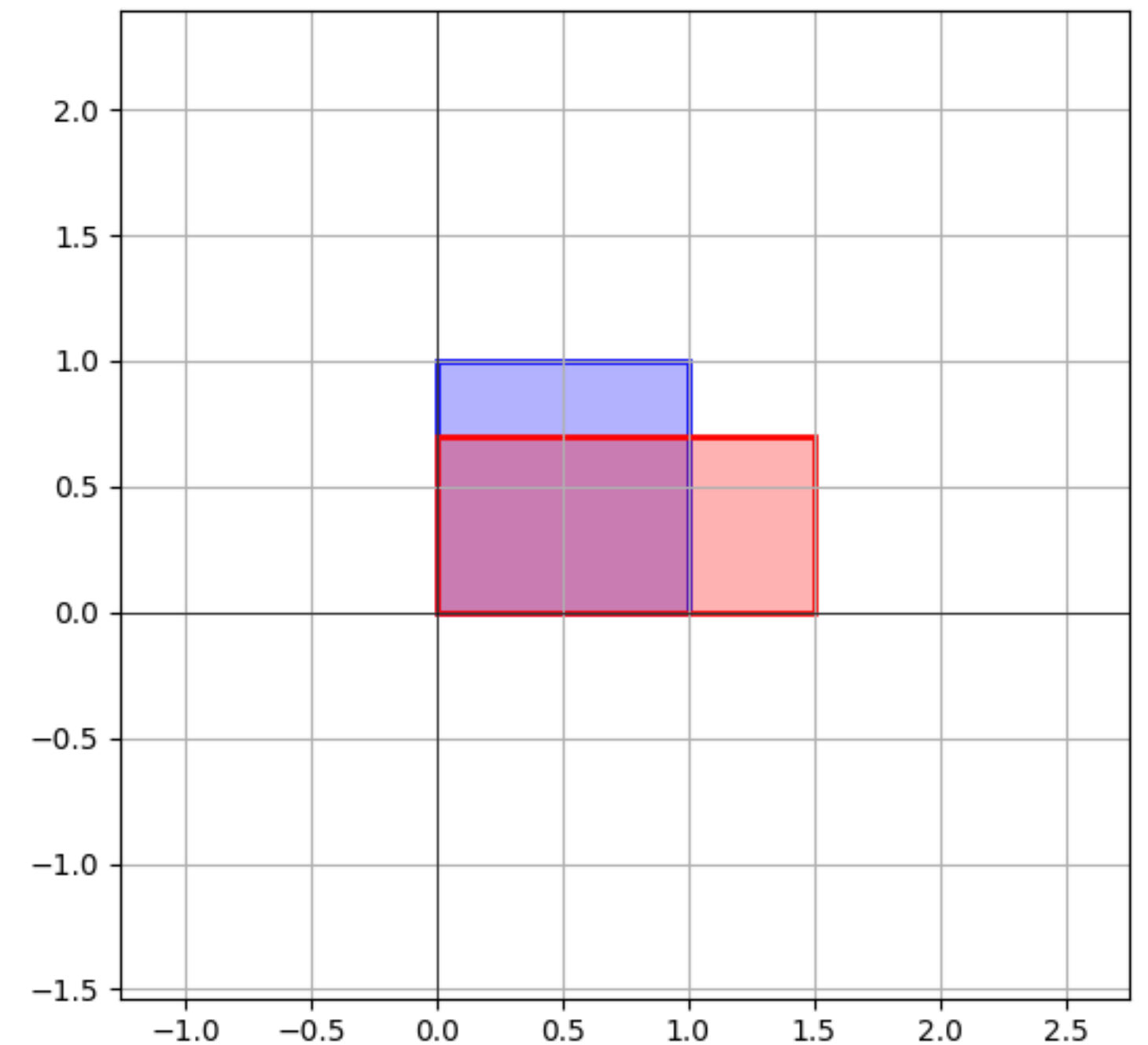
$$\begin{bmatrix} 1.5 & 0 \\ 0 & 0.7 \end{bmatrix}$$

$$\begin{bmatrix} 1.5 & 0 \\ 0 & 0.7 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1.5 \\ 0 \end{bmatrix} = 1.5 \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\lambda_1 = 1.5 \quad \vec{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1.5 & 0 \\ 0 & 0.7 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0.7 \end{bmatrix} = 0.7 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\lambda_2 = 0.7 \quad \vec{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



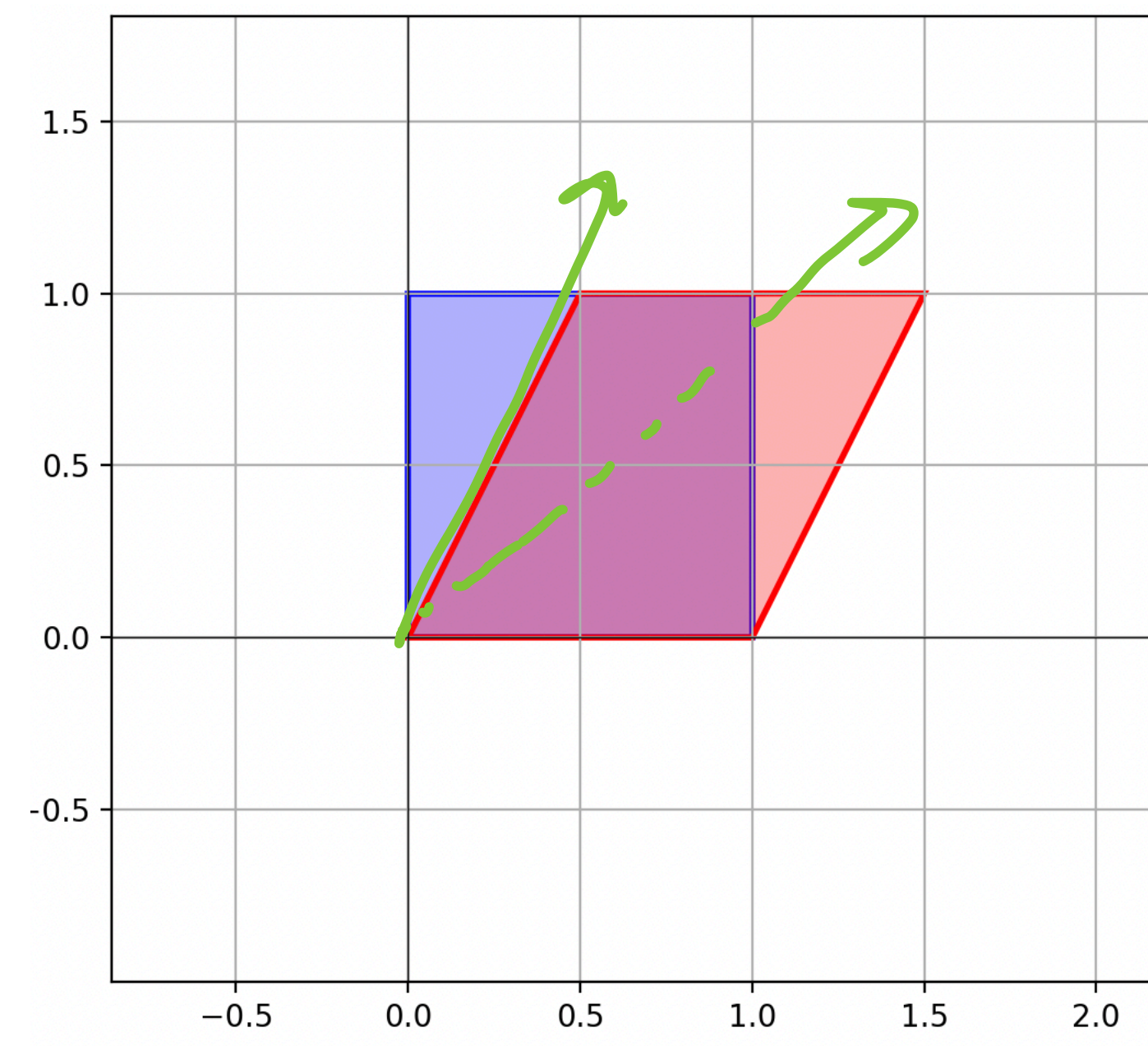
Example: Shearing

$$\begin{bmatrix} 1 & 0.5 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0.5 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 1 \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\lambda_1 = 1 \quad \vec{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0.5 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 0 \end{bmatrix} = 5 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 1 \begin{bmatrix} 5 \\ 0 \end{bmatrix}$$



Example

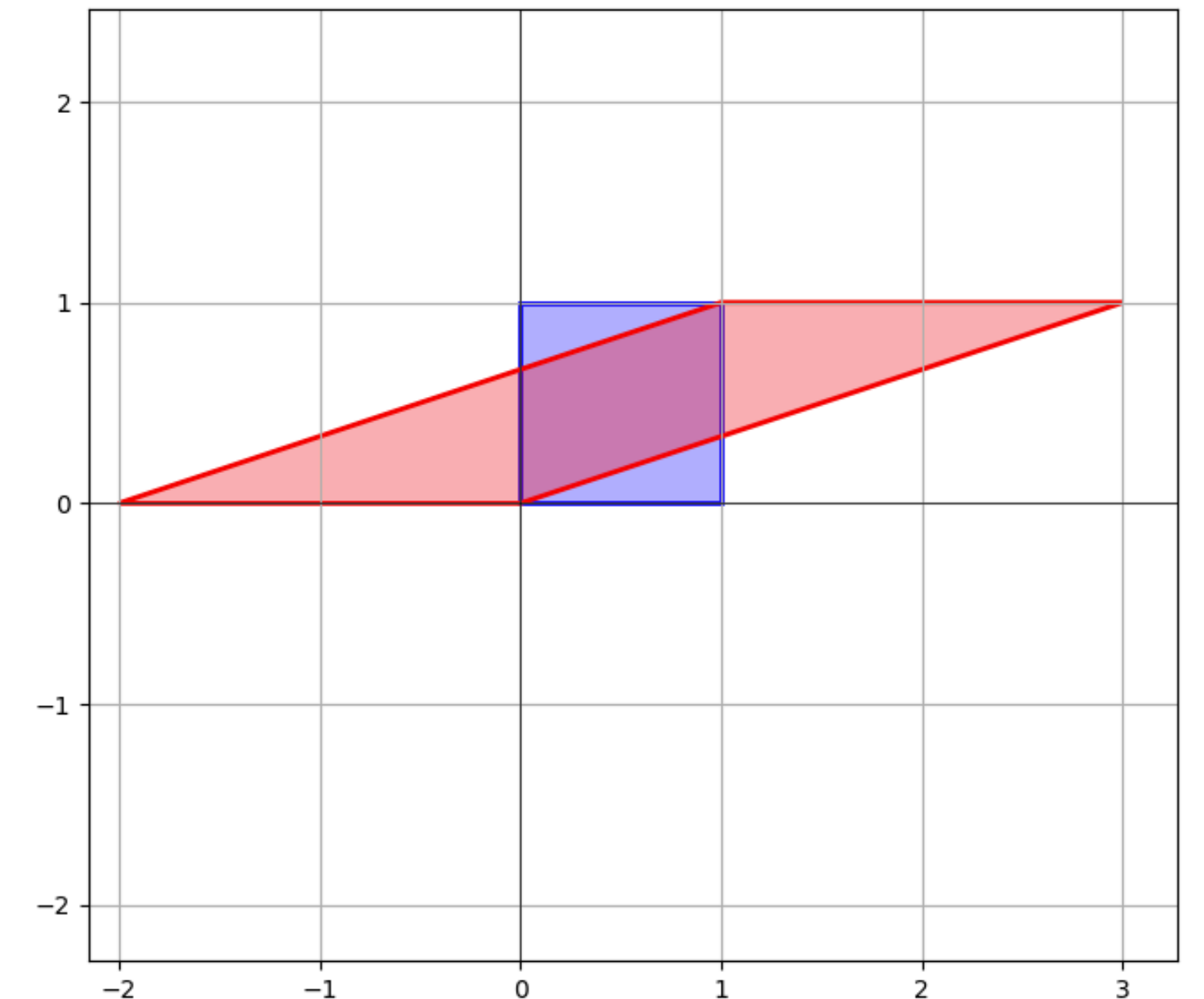
$$A = \begin{bmatrix} 3 & -2 \\ 1 & 0 \end{bmatrix} \quad \mathbf{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \mathbf{v} = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}$$

$$A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix} + \begin{bmatrix} -2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\lambda_2 = 1, \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$A \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix} + 0.5 \begin{bmatrix} -2 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\lambda_1 = 2, \quad \mathbf{v}_1 = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}$$



How do we verify eigenvalues
and eigenvectors?

Verifying Eigenvectors

Verifying Eigenvectors

Q. Determine if $\begin{bmatrix} 6 \\ -5 \end{bmatrix}$ or $\begin{bmatrix} 3 \\ -2 \end{bmatrix}$ are eigenvectors of $\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$ and determine their eigenvalues

Verifying Eigenvectors

Q. Determine if $\begin{bmatrix} 6 \\ -5 \end{bmatrix}$ or $\begin{bmatrix} 3 \\ -2 \end{bmatrix}$ are eigenvectors of $\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$ and determine their eigenvalues

A. Easy, work out the mv multiplication

matrix-vector

Verifying Eigenvalues

Verifying Eigenvalues

This is harder...

Verifying Eigenvalues

This is harder...

Q. Show that 7 is an eigenvalue of $\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$

Verifying Eigenvalues

This is harder...

Q. Show that 7 is an eigenvalue of $\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$

What vector do we check?

Verifying Eigenvalues

This is harder...

Q. Show that 7 is an eigenvalue of $\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$

What vector do we check?

Before we go over how to do this...

Verifying Eigenvalues (Warm Up)

Q. Verify that 1 is an eigenvalue of

$$B = \begin{bmatrix} 0.1 & 0.7 \\ 0.9 & 0.3 \end{bmatrix} \quad (\text{Hint. Recall Markov Chains})$$

A. $B \vec{v} = 1 \vec{v} = \vec{v}$

Steady-States and Eigenvectors

The Point: *Steady-state distributions are eigenvectors with eigenvalue 1*

$$A \vec{v} = \vec{v}$$

$$A \vec{v} - I \vec{v} = \vec{0}$$

$$(A - I) \vec{v} = \vec{0} \quad \leftarrow \text{solve}$$

$$v \in \text{Nul}(A - I)$$

v is a steady-state vector $\equiv$
 $\mathbf{v} \in \text{Nul}(A - I)^*$

*It must also be a probability vector

Verifying Eigenvalues

Q. Show that λ is an eigenvalue of A

A.

$$A \vec{v} = \lambda \vec{v}$$

$$A \vec{v} - \lambda \vec{v} = \vec{0}$$

$$A \vec{v} - \lambda I \vec{v} = \vec{0}$$

$$(A - \lambda I) \vec{v} = \vec{0}$$

solve
 $\vec{v} \in \text{Nul}(A - \lambda I)$

v is an eigenvector for $\lambda \equiv$
 $\mathbf{v} \in \text{Nul}(A - \lambda I)$

Verifying Eigenvalues

Q. Show that 7 is an eigenvalue of $\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$ ✓

$A - \lambda I$

A.
$$\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} - 7 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} - \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$$

$$= \begin{bmatrix} -6 & 6 \\ 5 & -5 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \quad \begin{array}{l} x_1 = x_2 \\ x_2 \text{ is free} \end{array}$$

$$\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix} + \begin{bmatrix} 6 \\ 2 \end{bmatrix} = \begin{bmatrix} 7 \\ 7 \end{bmatrix} = 7 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Problem

Verify that 2 is an eigenvalue of $\begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix}$

Answer

$$\begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix}$$

How many eigenvectors can
a matrix have?

Linear Independence of Eigenvectors

Thm. If $\mathbf{v}_1, \dots, \mathbf{v}_k$ are eigenvectors for distinct eigenvalues, then they are linearly independent

So an $n \times n$ matrix can have at most n eigenvalues

$$\begin{aligned} A \vec{v}_1 &= \lambda_1 \vec{v}_1 & \lambda_1 &\neq \lambda_2 & \alpha_1 \vec{v}_1 + \alpha_2 \vec{v}_2 &= \vec{0} & \alpha_2 (\lambda_2 - \lambda_1) \vec{v}_2 &= \vec{0} \\ A \vec{v}_2 &= \lambda_2 \vec{v}_2 & & & A \vec{0} &= \lambda_1 \vec{0} = \vec{0} & \text{blue } 0 & \text{green } \neq 0 \end{aligned}$$

$$\begin{aligned} A(\alpha_1 \vec{v}_1 + \alpha_2 \vec{v}_2) &= \alpha_1 A \vec{v}_1 + \alpha_2 A \vec{v}_2 = \alpha_1 \lambda_1 \vec{v}_1 + \alpha_2 \lambda_2 \vec{v}_2 \\ \lambda_1 (\dots) &= \alpha_1 \lambda_1 \vec{v}_1 + \alpha_2 \lambda_1 \vec{v}_2 \end{aligned}$$

Eigenspace

Fact. The set of eigenvectors for an eigenvalue λ of $A \in \mathbb{R}^{n \times n}$ form a subspace of $\mathbb{R}^n$

$$\text{Nul}(A - \lambda I)$$

is a subspace

$$A\vec{v} = \lambda\vec{v}$$

$$A\vec{u} = \lambda\vec{u}$$

$$A(\vec{v} + \vec{u}) =$$

$$A\vec{v} + A\vec{u} =$$

$$\lambda\vec{v} + \lambda\vec{u} =$$

$$\lambda(\vec{v} + \vec{u})$$

Eigenspace

Def. The set of eigenvectors for a eigenvalue λ of A is called the **eigenspace** of A for λ

It is the same as $\text{Nul}(A - \lambda I)$

How To: Basis of an Eigenspace

Q. Find a basis for the eigenspace of A corresponding to λ

A. Find a basis for $\text{Nul}(A - \lambda I)$

We know how to do this

$$\begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$0.5 \begin{bmatrix} x \\ y \end{bmatrix}$$

Example

$$\begin{bmatrix} -2 & 0 & 3 \\ 1 & 1 & -1 \\ -4 & 0 & 5 \end{bmatrix}$$

Determine a basis for the eigenspace corresponding to the eigenvalue 1

$$Nul(A - \lambda I)$$

$$\begin{bmatrix} -3 & 0 & 3 \\ 1 & 0 & -1 \\ -4 & 0 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} x_1 &= x_3 \\ x_2 &\text{ is free} \\ x_3 &\text{ is free} \end{aligned}$$

$$x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

How do we find
eigenvalues?

How do we find
eigenvalues?

We'll cover this next time...

Eigenvalues of Triangular Matrices

Thm. The eigenvalues of a triangular matrix are its entries along the diagonal

$$\begin{bmatrix} \boxed{1} & 0 & 0 \\ 2 & \boxed{4} & 0 \\ 3 & 5 & \textcolor{violet}{6} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 6 \end{bmatrix} = \textcolor{violet}{6} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\lambda_1 = 6 \quad \lambda_2 = 4 \quad \lambda_3 = 1$$

Example

$$\begin{bmatrix} 3 & 6 & -8 \\ 0 & 0 & 6 \\ 0 & 0 & 2 \end{bmatrix}$$

Determine the eigenvectors and values of the above matrix

Linear Dynamical Systems

Recall: Linear Dynamical Systems

Recall: Linear Dynamical Systems

Def. A (discrete time) linear dynamical system is a described a $n \times n$ matrix A . It's evolution function is the matrix transformation $\mathbf{x} \mapsto A\mathbf{x}$

Recall: Linear Dynamical Systems

Def. A (discrete time) linear dynamical system is a described a $n \times n$ matrix A . It's **evolution function** is the matrix transformation $\mathbf{x} \mapsto A\mathbf{x}$

The **possible states** of the system are vectors in $\mathbb{R}^n$

Recall: Linear Dynamical Systems

Def. A (discrete time) linear dynamical system is a described a $n \times n$ matrix A . It's **evolution function** is the matrix transformation $\mathbf{x} \mapsto A\mathbf{x}$

The **possible states** of the system are vectors in $\mathbb{R}^n$

Given an **initial state vector** $\mathbf{v}_0$, we can determine the **state vector** of the system after i time steps:

$$\mathbf{v}_i = A\mathbf{v}_{i-1} \quad (\text{linear recurrence relation})$$

Recall: State Vectors

$$\mathbf{v}_0$$

$$\mathbf{v}_1 = A\mathbf{v}_0$$

$$\mathbf{v}_2 = A\mathbf{v}_1 = A(A\mathbf{v}_0)$$

$$\mathbf{v}_3 = A\mathbf{v}_2 = A(AA\mathbf{v}_0)$$

$$\mathbf{v}_4 = A\mathbf{v}_3 = A(AAA\mathbf{v}_0)$$

$$\mathbf{v}_5 = A\mathbf{v}_4 = A(AAAA\mathbf{v}_0)$$

$$\vdots$$

The **state vector** $\mathbf{v}_k$ is the state of the system after k time steps

The Issue

The Issue

The eq. $\mathbf{v}_k = A^k \mathbf{v}_0$ doesn't tell us much about $\mathbf{v}_k$

The Issue

The eq. $\mathbf{v}_k = A^k \mathbf{v}_0$ doesn't tell us much about $\mathbf{v}_k$

It's defined in terms of A itself, which
doesn't say much about how the system behaves

The Issue

The eq. $\mathbf{v}_k = A^k \mathbf{v}_0$ doesn't tell us much about $\mathbf{v}_k$

It's defined in terms of A itself, which doesn't say much about how the system behaves

It's also difficult computationally because matrix multiplication is expensive

(Closed-Form) Solutions

(Closed-Form) Solutions

A **(closed-form) solution** of a linear recurrence relation $\mathbf{v}_{i+1} = A\mathbf{v}_i$ is an expression for $\mathbf{v}_k$ that does *not* contain A^k or previously defined terms

(Closed-Form) Solutions

A **(closed-form) solution** of a linear recurrence relation $\mathbf{v}_{i+1} = A\mathbf{v}_i$ is an expression for $\mathbf{v}_k$ that does *not* contain A^k or previously defined terms

In other word, it doesn't depend on A^k and isn't **recursive**

Example

$$\mathbf{v}_k = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \mathbf{v}_{k-1} \quad \mathbf{v}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \mathbf{v}_1$$

$$\mathbf{v}_k = \begin{bmatrix} k+1 \\ k \end{bmatrix} \quad k \geq 0$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

Solutions with Eigenvectors as Initial States

Solutions with Eigenvectors as Initial States

It's easy to give a closed-form solution if the initial state is an eigenvector

$$\mathbf{v}_k = A^k \mathbf{v}_0 = \lambda^k \mathbf{v}_0$$

Solutions with Eigenvectors as Initial States

It's easy to give a closed-form solution if the initial state is an eigenvector

$$\mathbf{v}_k = A^k \mathbf{v}_0 = \lambda^k \mathbf{v}_0$$

No dependence on A^k or $\mathbf{v}_{k-1}$

Solutions with Eigenvectors as Initial States

It's easy to give a closed-form solution if the initial state is an eigenvector

$$\mathbf{v}_k = A^k \mathbf{v}_0 = \lambda^k \mathbf{v}_0$$

No dependence on A^k or $\mathbf{v}_{k-1}$

The Key Point. *This is still true of linear combinations of eigenvectors*

Solutions in terms of eigenvectors

Let's simplify $A^k \mathbf{v}$, given we have eigenvectors $\mathbf{b}_1, \mathbf{b}_2$ for A which span all of $\mathbb{R}^2$:

$$A^k (\mathbf{b}_1 + \mathbf{b}_2) =$$

$$A^k \mathbf{b}_1 + A^k \mathbf{b}_2 =$$

$$\lambda_1^k \mathbf{b}_1 + \lambda_2^k \mathbf{b}_2 =$$

Eigenvectors and Growth in the Limit

Thm. For a linear dynamical system A with initial state $\mathbf{v}_0$, if $\mathbf{v}_0$ can be written in terms of eigenvectors $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_k$ of A with eigenvalues $\lambda_1 > \lambda_2 \dots \geq \lambda_k$ then $\mathbf{v}_k \sim \lambda_1^k c_1 \mathbf{b}_1$ for some constant c_1

In other words, in the long term,
the system grows exponentially in
the largest eigenvalue

Eigenbases

Eigenbases

Def. An **eigenbasis** of $\mathbb{R}^n$ for $A \in \mathbb{R}^{n \times n}$ is a basis of $\mathbb{R}^n$ made up entirely of eigenvectors of A

Eigenbases

Def. An **eigenbasis** of $\mathbb{R}^n$ for $A \in \mathbb{R}^{n \times n}$ is a basis of $\mathbb{R}^n$ made up entirely of eigenvectors of A

*We can represent vectors as **unique** linear combinations of eigenvectors*

Eigenbases

Def. An **eigenbasis** of $\mathbb{R}^n$ for $A \in \mathbb{R}^{n \times n}$ is a basis of $\mathbb{R}^n$ made up entirely of eigenvectors of A

*We can represent vectors as **unique** linear combinations of eigenvectors*

Not all matrices have eigenbases

Eigenbases and Growth in the Limit

Thm. For a linear dynamical system A with initial state $\mathbf{v}_0$, if A has an eigenbasis $\mathbf{b}_1, \dots, \mathbf{b}_k$, then

$$\mathbf{v}_k \sim \lambda_1^k c_1 \mathbf{b}_1$$

for some constant c_1 , where λ_1 is the **largest eigenvalue** of A and $\mathbf{b}_1$ is its **eigenvector**

Eigenbases and Growth in the Limit

Thm. For a linear dynamical system A with initial state $\mathbf{v}_0$, if A has an eigenbasis $\mathbf{b}_1, \dots, \mathbf{b}_k$, then

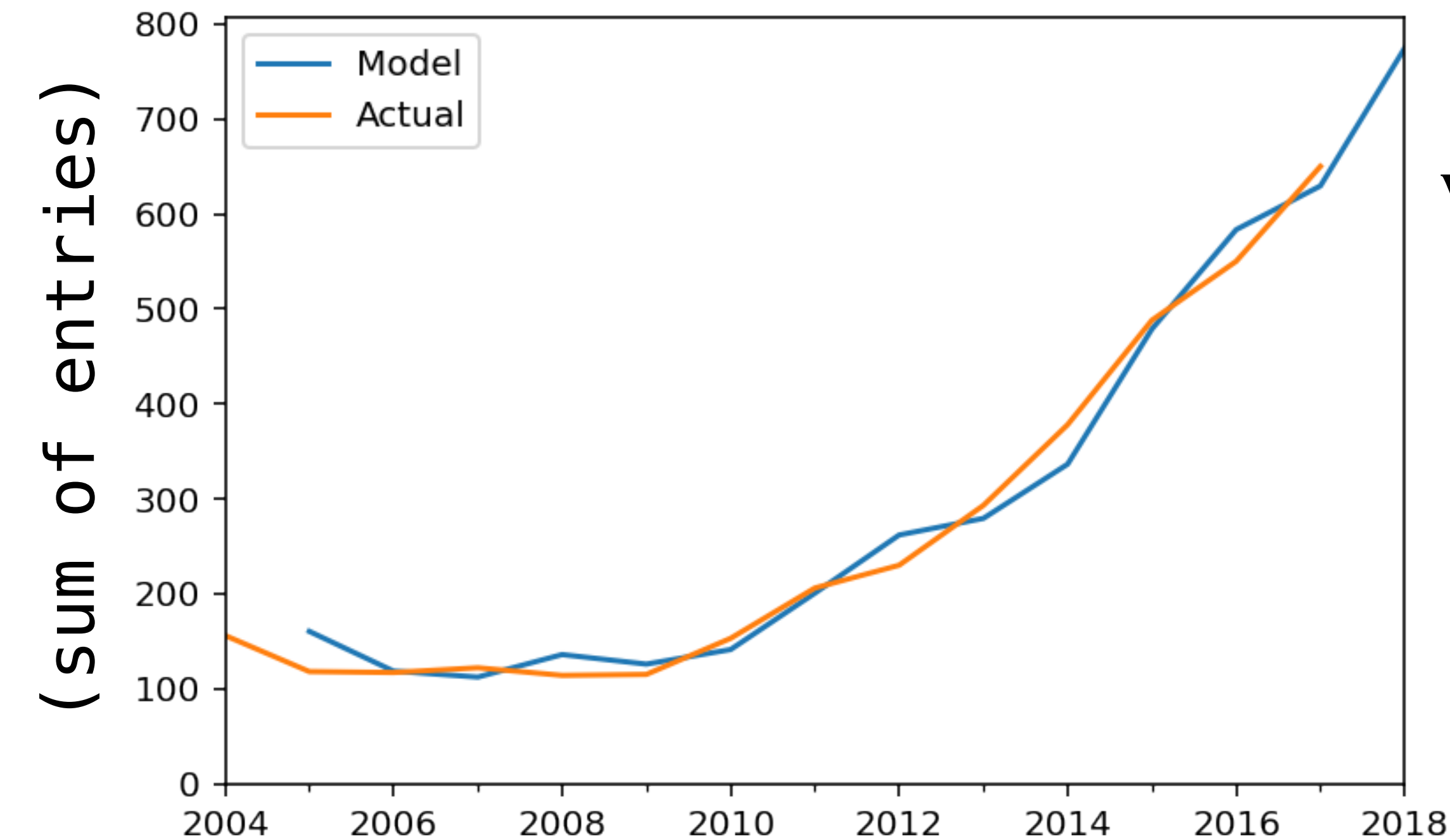
$$\mathbf{v}_k \sim \lambda_1^k c_1 \mathbf{b}_1$$

for some constant c_1 , where λ_1 is the **largest eigenvalue** of A and $\mathbf{b}_1$ is its **eigenvector**

The largest eigenvalue describes the long-term exponential behavior of the system

Example: CS Major Growth

see the notes for more details



$$\mathbf{v}_0 = \begin{bmatrix} v_{0,1} \\ v_{0,2} \\ v_{0,3} \\ v_{0,4} \end{bmatrix} = \begin{bmatrix} \# \text{ of year 1 students enrolled in 2024} \\ \# \text{ of year 2 students enrolled in 2024} \\ \# \text{ of year 3 students enrolled in 2024} \\ \# \text{ of year 4 students enrolled in 2024} \end{bmatrix}$$

$$\mathbf{v}_k = A^k \mathbf{v}_0$$

(A is determined by least squares)

This is clearly exponential. If we want to "extract" the exponent, we need to look at the *largest eigenvalue*

Another Example: Golden Ratio

A Special Linear Dynamical System

$$\mathbf{v}_{k+1} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \mathbf{v}_k \quad \mathbf{v}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

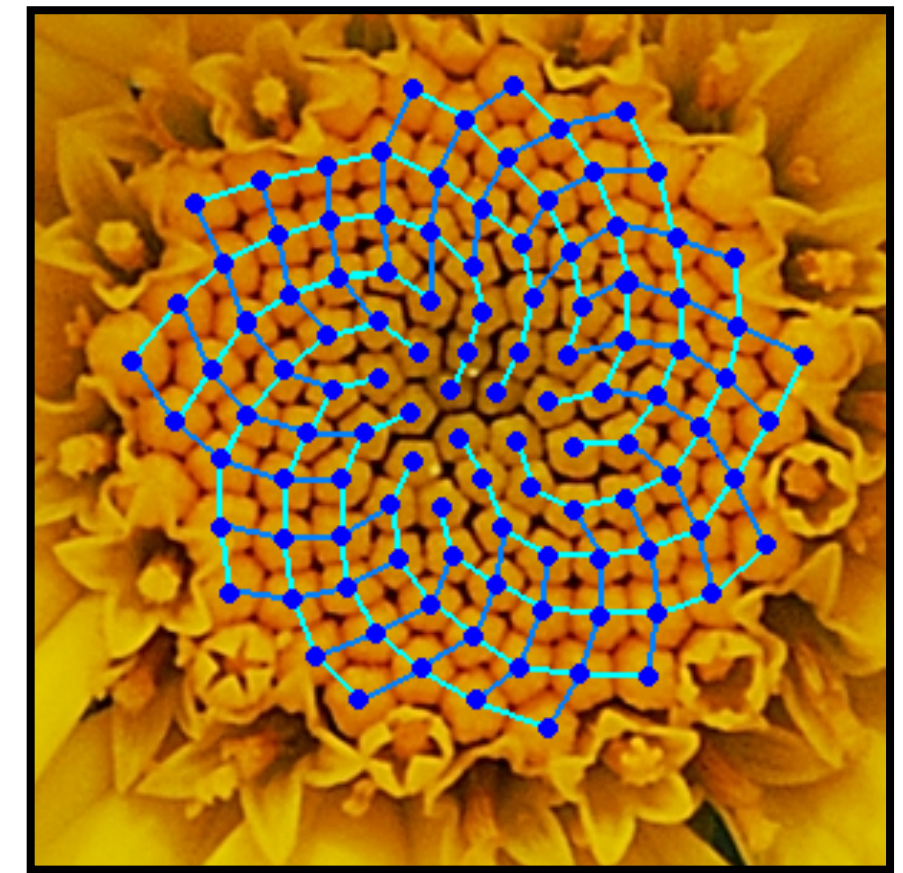
Fibonacci Numbers

$$F_0 = 0$$

$$F_1 = 1$$

$$F_k = F_{k-1} + F_{k-2}$$

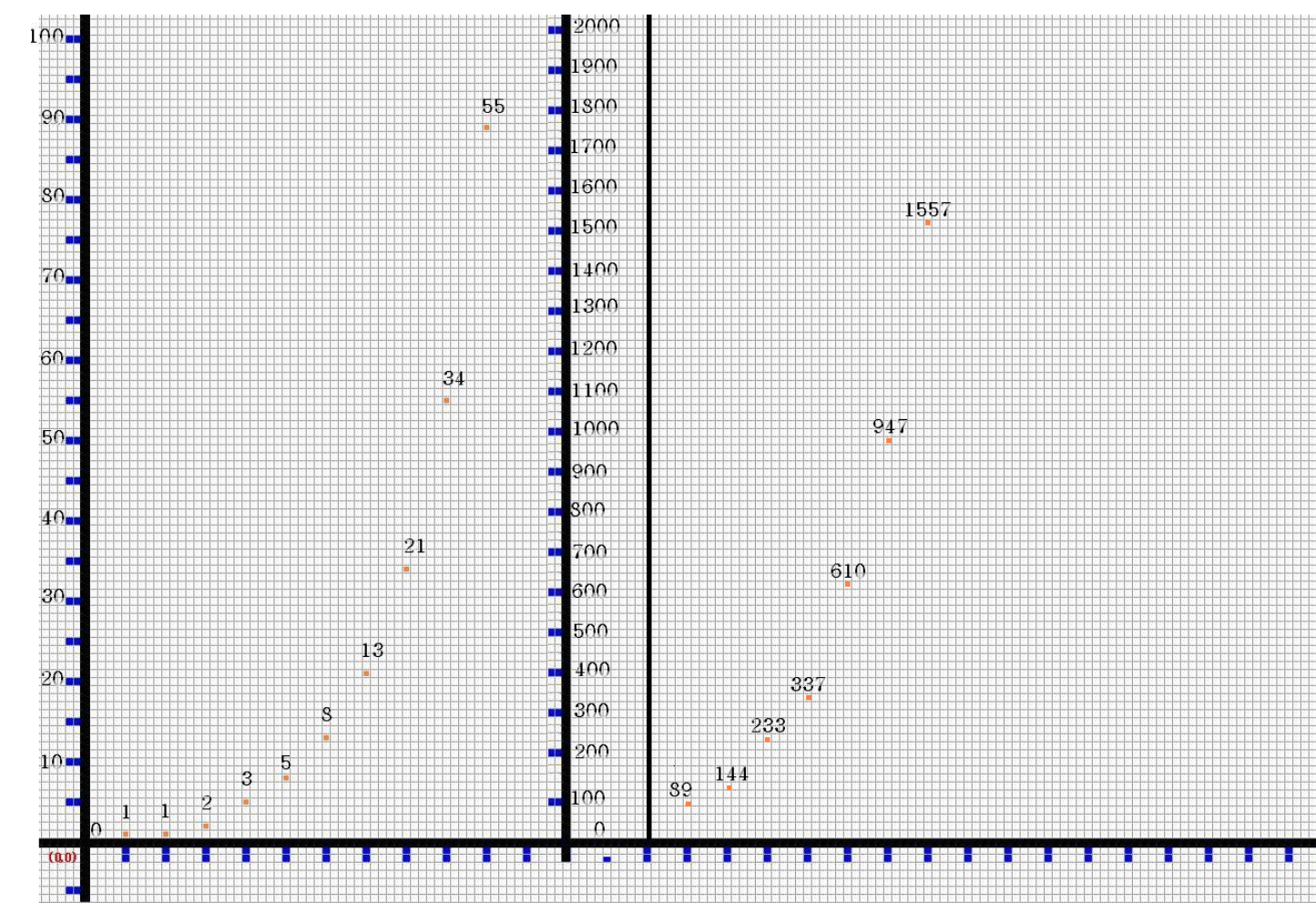
```
define fib(n):  
  curr, next ← 0, 1  
  repeat n times:  
    curr, next ← next, curr + next  
  return curr
```



The Fibonacci numbers are defined in terms of a recurrence relation

Golden Ratio

$$\varphi = \frac{1 + \sqrt{5}}{2}$$



The "long term behavior" is the Fibonacci sequence is defined by the golden ratio: $F_k \sim \varphi^k$

φ is an eigenvalue of $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$